Strata Cobot: Autonomous Ore Sorting and Haulage with a Manipulator and a Mobile Robot in ROS 2

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***Abstract*—We describe a course project on cooperative autonomy between two simulated robots in Gazebo: a fixed UR7e arm and a skid-steer ground vehicle, the eBot. The arm looks at a conveyor through a depth camera, lifts every ore block with an electromagnet and deposits it in the bin that matches its mineral. The vehicle computes routes around scattered boulders from a static occupancy map, tracks them, and picks up an ore crate during its circuit. No motion-planning or navigation framework (for example MoveIt 2 or Nav2) is involved; the detector, the kinematic solver, the arm and vehicle planners and both controllers were implemented by the team. Natural ore is seldom uniformly coloured, so blocks are segmented geometrically from depth and labelled by the proportions of mineral hues across the entire top surface. This labelling survives impurities over half of the coloured area and identifies 70% of unseen photographs of real specimens, compared with 11% for narrow hue thresholds. The vehicle further delivers its crate to the arm and computes a fresh route when it meets a boulder absent from the map. Localisation error is 2–3 mm, a kinematic solution takes 0.1–0.4 s, and route lengths equal those of a reference shortest route. In every repeated trial all ores reached the right bin and every circuit finished, without collisions or protective stops.**

***Index Terms*—ROS 2, mobile manipulation, inverse kinematics, A* path planning, pure pursuit, RGB-D perception**

I. Introduction

The scenario is an off-world mine operated without people. A wheeled vehicle travels across boulder-strewn ground to a dig site, receives ore and brings it back to a processing station. At the station a fixed arm inspects every block with a camera and assigns it to one of three bins: azurite (blue), malachite (green) or vanadinite (orange). The e-Yantra Strata Cobot environment [1] provides the world model, both robots and their sensors.

Two problems are addressed. First, the arm has to locate four blocks on a conveyor, lift each one magnetically and release it above the correct bin. Second, the vehicle has to complete a circuit of three legs between prescribed poses among boulders, collecting a crate along the way, using routes it computes on its own. We rely on no external planning or navigation package; every planner and controller described here was written for this work.

Our contributions are:

- a depth-based block detector paired with a hue-proportion classifier that tolerates impure ore, localising blocks to 1.7–2.7 mm;
- an arm pipeline combining damped least-squares inverse kinematics with resolved-rate descents and force-checked grasps;
- a grid planner that achieves the reference shortest length and a pure-pursuit tracker that holds the vehicle within 0.09 m of the route;
- repeated randomised trials evaluating both robots.

II. System Overview

Software runs under Ubuntu 24.04 using ROS 2 [7] Jazzy and Gazebo Harmonic, split into four Python nodes with two per robot (Fig. 1). Since the arm takes only joint- or Cartesian-velocity commands and the vehicle takes only twists on `/cmd_vel`, each node runs its own feedback loop at 20–50 Hz.

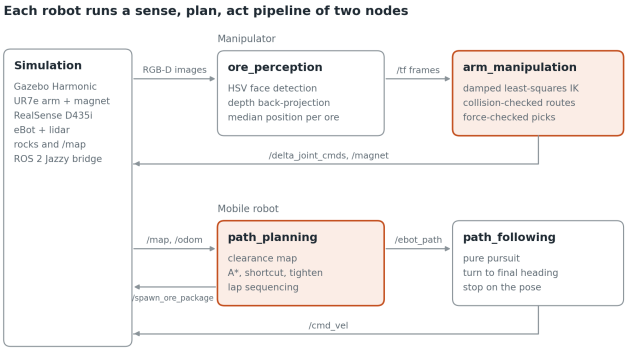


Fig. 1. How the nodes connect. For readability, feedback topics are left out (`/tcp_pose_raw`, `/joint_states`, tool force and `/arm_status` on the arm side; `/odom` on the vehicle side).

An electromagnet is mounted on the UR7e flange, and a RealSense D435i is fixed to the arm base, pointing down at the conveyor. The servo interface limits every joint to 0.60 rad/s and latches a protective stop on joint-limit violations, singularities, contact and overload. The vehicle carries a planar lidar, and a static occupancy grid of the arena arrives once on `/map`.

III. Perception

The blocks measure $101.6 \times 101.6 \times 76.2$ mm and are black apart from a tinted upper face. Our initial detector applied narrow HSV bands per mineral and accepted blobs of suitable area and aspect. It works only while each face shows a

single clean, saturated hue. Genuine specimens break such bands through intergrown minerals, dark or rust-coloured host rock, dust and varying illumination. The final design therefore decouples localisation from identification.

A. Geometric segmentation

For each pixel (u, v) in the depth image registered to colour, with depth z, the camera intrinsics give the 3-D point

$$\mathbf{p}_c = \left(\frac{(u - c_x)z}{f_x}, \frac{(v - c_y)z}{f_y}, z \right), \tag{1}$$

which is then expressed in the arm's `base_link` frame. Nothing on the conveyor except a block's upper face lies between 66 and 86 mm above the belt, so the points in that slab, grouped into connected components, form the candidates; the slab also keeps adjacent blocks apart, since their side walls fall below it. A component survives only if its fitted rectangle has sides of 80–130 mm, and the rectangle's centre is taken as the block position. An earlier median estimate was biased 4–9 mm towards the camera by boundary pixels whose depth blends with the floor. Compared with simulator ground truth, the final estimates deviate by 1.7–2.7 mm.

B. Hue-proportion labelling

Identification uses every pixel of the face. For each mineral we compute the fraction of face pixels whose hue lies in its band (blue 95–130, green 40–90, orange-red 0–20 and 170–180, OpenCV scale 0–180), subject to saturation and brightness minima that reject only grey, black and white host rock ($S \geq 80$; $V \geq 15$ for blue, because natural azurite is very dark; $V \geq 40$ for green). Orange must also satisfy $S \geq 140$ and $V \geq 100$, so rusty brown stone is not mistaken for vanadinite. The leading mineral is accepted when its fraction is at least 3% and at least double that of every competitor; otherwise the face is labelled *uncertain*. Contamination thus reduces the leading fraction rather than destroying the detection. Across the 20 frames gathered before labelling, faces within 5 cm of each other are merged as one block whose label is the majority result; a block that never obtains a clear result is reported rather than guessed. Blocks of each mineral are numbered from 1 starting nearest the arm and broadcast as tf2 frames at 10 Hz, located at the median of the latest 30 estimates (Fig. 2).

C. Synthetic contamination

Because the simulator's blocks are perfectly clean, contamination was added artificially to a recorded camera image: random patches of dark stone (40%), brown host rock (30%), white quartz (15%) and streaks of the two other minerals (15%), covering 0–70% of each tinted face, combined with random brightness scaling (×0.6–1.3), per-channel tint (±8%) and noise. Fig. 3 reports 40 trials with four blocks per contamination level. Under dim lighting the narrow-band detector misses 3% of clean blocks, and at 20% contamination it recovers fewer than half. The hue-proportion classifier labels all blocks correctly up to 50% contamination and 84% of them at 70%, and never outputs the wrong mineral; undecidable faces are flagged uncertain, so no block is placed in an incorrect bin.

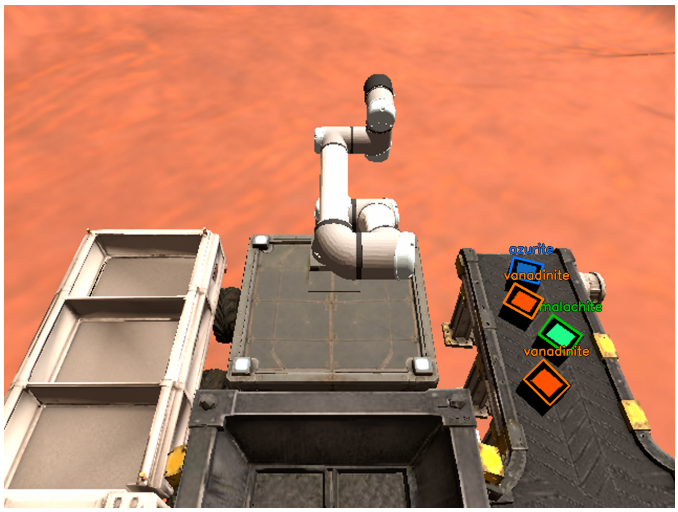


Fig. 2. Colour image from the RealSense with the four blocks found on the conveyor (right); the three bins on the cargo rover appear at the left.

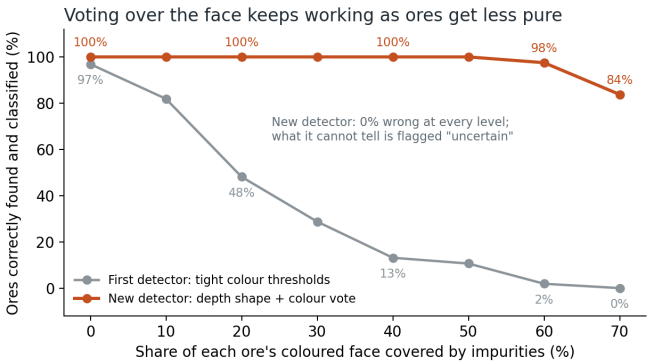


Fig. 3. Fraction of blocks located and labelled correctly versus the portion of the tinted face hidden by synthetic contamination (160 blocks per level).

D. Photographs of real specimens

Synthetic patches remain artificial, so both classifiers were also tested on 180 specimen photographs from Wikimedia Commons [6]: 60 per mineral, many showing intergrowths and host rock. Lacking depth, each photograph is segmented by Lab colour distance from the median colour of its border. Brightness minima and acceptance ratios were set by grid search over the first 30 images per mineral, maximising correct labels minus twice the wrong ones, and results are given for the remaining 90 (Table I).

Narrow bands tuned on clean simulated faces identify 11% of the real specimens, against 70% for the hue-proportion classifier. Using the simulator-derived limits unchanged gave 57% correct with 2% wrong; calibration therefore accepts a few extra errors in exchange for far fewer uncertain labels. Azurite is the most difficult case: its blue is very dark, and

TABLE I
CLASSIFICATION OF 90 HELD-OUT REAL PHOTOGRAPHS

Mineral	First detector	Voting detector		
		Correct	Wrong	Uncertain
Azurite	10%	37%	17%	47%
Malachite	7%	80%	7%	13%
Vanadinite	17%	93%	0%	7%
All	**11%**	**70%**	**8%**	**22%**

many samples carry a green malachite crust, which the two-to-one criterion correctly declines to resolve. Once physical samples are available, a learned image classifier such as a compact convolutional network would be the logical successor.

IV. Manipulation

A. Workspace and bins

Bin locations are not supplied, so they were recovered from the GLB collision model of the rover carrying the bins by mapping its vertices into `base_link`. Each bin's opening is 0.40 m long in y and 0.26–0.31 m wide in x. The vanadinite bin's centre is 0.93 m from the base, out of reach with a vertical tool, so blocks are released at the point of each bin closest to the arm, 8 cm inside the walls (Fig. 4).

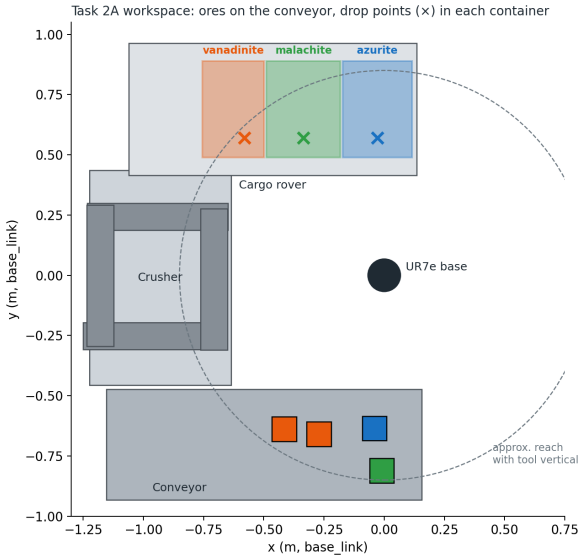


Fig. 4. Plan view of the arm's surroundings in `base_link`: obstacles represented as collision boxes, blocks on the conveyor, and the release point (×) chosen in every bin.

B. Inverse kinematics

The kinematic model, together with a sampled point cloud of every link's collision mesh, is parsed from `/robot_description`. The magnet is circular, so yaw about the vertical is unconstrained and the solver enforces five conditions: tool position $\mathbf{p}$ and a downward tool axis $\mathbf{z}$. Defining $\mathbf{e} = [\mathbf{p}^{*} - \mathbf{p};\ (\mathbf{z} \times -\hat{\mathbf{k}})_{xy}]$ and taking the matching 5×6 rows J of a finite-difference Jacobian, every iteration performs the damped least-squares update [4]

$$\Delta\mathbf{q} = J^{T}\left(JJ^{T} + \lambda^{2} I\right)^{-1}\mathbf{e}, \qquad \lambda = 0.02, \tag{2}$$

limited to 0.3 rad per step. A converged configuration is kept only when it respects the joint limits by 0.3 rad, stays 3 cm away from all obstacle boxes, holds the tool within 1° of vertical and keeps the minimum singular value of J above 0.04. Accepted configurations are cached, and those closest to a new goal serve as additional starting guesses; without them, solves begun above the bins could not return to the conveyor. A first implementation based on constrained SLSQP needed minutes per plan, whereas (2) solves in 0.1–0.4 s.

C. Motion and grasping

Long moves interpolate linearly in joint space and are accepted when every intermediate configuration has 2.5 cm of clearance. Failing that, the planner attempts a raise–cross–lower detour at 0.30–0.50 m height, then a path via a staging pose above open floor. A carried block is appended to the collision model beneath the tool. Joint goals are reached by a proportional controller limited to 0.30 rad/s.

With an 81 mm magnet disc and a 101.6 mm square face, the tool must land within roughly 10 mm of the centre and flat on the face. Vertical motions therefore use resolved-rate control [5]: a bounded tool velocity $\mathbf{v}$ proportional to position error and a tilt-correcting rate ω become joint rates through $\dot{\mathbf{q}} = J^{+}[\mathbf{v}; \omega]$. For each grasp the tool pauses 10 cm over the face, energises the magnet, moves down at 4 cm/s and then 1.2 cm/s, and halts on reaching the face or when contact force passes 2.5 N. Once raised, the tool force with its own weight subtracted is near 1.47 N when a 0.15 kg block is attached; readings under 0.8 N trigger up to three retries. When a protective stop latches, the node clears it, retreats upwards and replans the step it was executing.

V. Path Planning and Following

A. Task

The vehicle travels from home $(0, 0)$ to the ore pickup pose $(6.0, 1.9)$, requests the crate, proceeds to the arm pose $(0, 1.9)$ and goes back home. A route is admissible when it begins and ends precisely at these poses and enters no occupied cell; the vehicle must remain within 0.5 m of it and settle within 0.3 m and 0.15 rad of each pose. Route quality is scored by

$$\eta = \min\left(1,\ \frac{L_{\mathrm{ref}}}{L_{\mathrm{path}}}\right), \tag{3}$$

with L_{ref} the shortest route for a 0.395 m disc that encloses the 0.62×0.49 m body.

B. Planner

Applying a Euclidean distance transform to the 5 mm `/map` (1784×1380 cells) yields each cell's distance to the closest boulder or wall. On a 5 cm lattice sampled at cell centres, 8-connected A* [2] searches the cells whose distance satisfies a clearance level c. Close to the fixed start and goal poses

this requirement is relaxed to 0.30 m, because the pickup pose sits just 0.375 m from the boundary wall. The A* result is then shortcut, linking each vertex to the farthest subsequent vertex reachable in a straight line at clearance c, and tightened, moving every corner towards the segment joining its neighbours and halving the move until both adjacent segments remain clear. All straight segments are confirmed on the full-resolution map every 5 mm. Tightening eliminates lattice artefacts, so each route hugs boulders at precisely the requested clearance (Fig. 5).

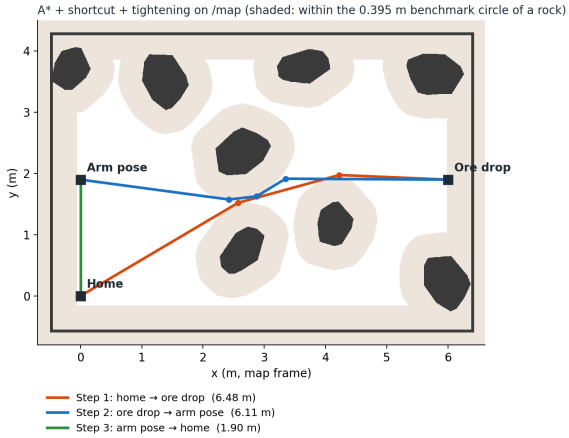


Fig. 5. Routes computed for the circuit's three legs; shading marks cells closer than 0.395 m to a boulder or wall.

C. Selecting the clearance

Although the reference models the vehicle as a 0.395 m disc, passing a boulder in a straight line only requires the 0.245 m half-width to clear it. Fig. 6 plots this compromise for one layout: up to roughly 0.40 m the circuit length barely changes, but beyond it tight passages close and the circuit lengthens by as much as 13%. The planner therefore first plans at 0.395 m to approximate the reference, then tests levels from 0.40 down to 0.34 m and keeps the largest one whose length stays within 1 cm of that approximation. At 0.38 m this leaves 13.5 cm on each side, whereas the worst measured tracking error is 9 cm.

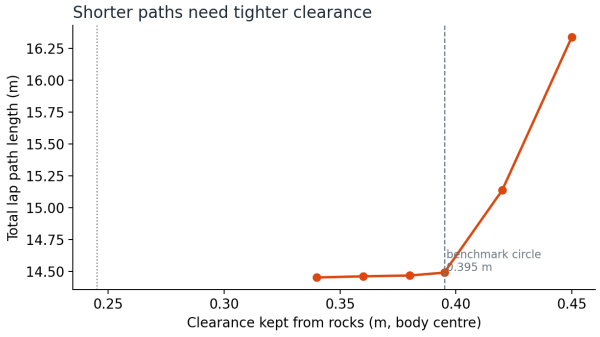


Fig. 6. Circuit length as a function of the clearance held from boulders for a single layout; above 0.45 m no route exists.

D. Tracker

Route tracking uses pure pursuit [3] with 0.30 m lookahead at 0.35 m/s. If heading error exceeds 40° the vehicle rotates in place; near the route's end it decelerates, rotates to the final heading and halts. During normal driving the lidar only reduces speed near obstacles. The planner monitors `/odom`, moves on to the next leg once the vehicle has been stationary on the goal pose for 0.5 s, and calls `/spawn_ore_package` asynchronously at the pickup pose.

VI. Integration: Hand-off and Re-planning

A. Crate transfer from vehicle to arm

For the complete mission the arm is inserted into the vehicle's world. With transfer enabled, the planner pauses after the second leg at the arm pose, raises `/handoff/ready` and waits. A transfer node, reusing the sorting node's arm model, planner and controllers, takes control. It predicts the crate's position from the vehicle's odometry (which agrees with simulator ground truth to within a millimetre), then corrects the height and, if the full face is visible, the centre using the depth camera, reaching 4 mm accuracy in tests. The vehicle body is inserted into the arm's collision boxes. The crate ($177.8 \times 139.7 \times 76.2$ mm, 0.30 kg) is lifted magnetically, verified by a 2.94 N force reading, raised vertically until 7 cm above the deck, transported with its volume in the collision model, set down on the conveyor's input end and released. The node finally sets `/handoff/done`, and the vehicle returns home; a complete mission lasts roughly 2 minutes.

B. Responding to unmapped obstacles

Since every route stays at least 0.30 m from mapped boulders, any lidar point within 0.26 m of the upcoming 1.6 m of route must belong to something unmapped, and is also close enough to strike the 0.245 m half-width body. Following three successive detections, the tracker halts and sends the nearby lidar points on `/ebot_obstacles`. The planner inserts them into its occupancy data, recomputes the distance transform and produces a replacement route from the vehicle's current pose in about 0.1 s. In a test where a box was spawned onto the route during a leg, the vehicle halted, drove the replacement route and finished the circuit, keeping at least 1.16 m from the box centre.

VII. Results

Every trial began from a newly launched world with random ore placement or boulder layout and was scored by the automatic evaluator bundled with the simulator. Table II lists the outcomes.

In all vehicle trials the crate was collected with no collision, tracking error never exceeded 0.09 m, and the vehicle settled within 0.15 m and 0.04 rad of every pose. All grasps registered 1.46–1.47 N. The slight deficit in trial Robot 3 stems from the reference: for an identical layout, the reference second leg was 6.14 m in one run and 6.07 m in the next, while our route remained 6.11 m. Recovering that margin consistently would need clearances under 0.34 m, with a genuine collision risk.

TABLE II
TRIAL RESULTS

Trial	Outcome
Arm 1	All four blocks found, grasped and binned correctly; no protective stops
Arm 2	New mineral mix with one block 0.83 m from the base: all four binned correctly in 167 s; no protective stops
Arm 3	Voting detector (simulator limits): 4/4 classified in 20/20 frames, sorted correctly; 0 safety stops
Arm 4	Voting detector (calibrated limits), two azurite ores: 4/4 sorted correctly in 180 s; 0 safety stops
Robot 1	Initial planner: circuit of 14.74 m against a 14.46 m reference, $\eta = 0.98$
Robot 2	Final planner: 14.55 m against 14.58 m, $\eta = 1.00$
Robot 3	Final planner: 14.48 m against 14.43 m, $\eta = 0.996$

VIII. Challenges and Lessons

Table III lists the principal issues encountered; most surfaced only once both robots were exercised end to end under automatic scoring.

TABLE III
MAIN PROBLEMS AND FIXES

Problem	Cause and fix
No path found	Drop pose 0.375 m from the wall, plus clearance lost to map downsampling. Fixed by full-resolution clearance and a relaxed limit near fixed poses.
Paths 2% long	Coarse-grid checks and grid-snapped corners. Fixed by fine-map checks, tightening and reference-aware clearance.
IK slow or stuck	SLSQP evaluated all checks in every gradient step; poor seeds over the containers. Fixed by DLS IK and a seed library.
Vanadinite unreachable	Container centre 0.93 m away. Fixed by releasing at the nearest reachable point inside it.
Ores never found	tf frame list writes a space after each name. Fixed by a regular-expression parse.

We draw three lessons. Early end-to-end runs against an automatic scorer revealed losses that offline plots concealed. Measuring the scene geometry directly uncovered both the wall next to the pickup pose and the out-of-reach bin centre. Finally, a solver tailored to the task, with validity checks run once on its output, proved roughly 300 times faster than a generic constrained optimiser.

IX. Conclusion

In simulation, both robots accomplish their tasks end to end with planners and controllers developed solely for this project: all ores were binned correctly and all circuits completed collision-free. The robots also cooperate, with the vehicle transferring its crate to the arm and rerouting around boulders missing from the map. Planned extensions are a learned classifier trained on physical ore samples, quicker sorting through blended motions and pick ordering, and deployment on real hardware, where illumination and sensor noise will stress the hue limits and the force-based grasp check.

Acknowledgment

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